

박수민_0618_PWM_제어

```
#main.c
#define F_CPU 16000000UL
#include <avr/io.h>
#include <avr/interrupt.h> // sei, ISR 등
#include <util/delay.h>
#include <stdio.h>

#define LED_TIME 20

void turn_on_LED_in_PWM_manner(int dim)
{
    int i;

    PORTA = 0xFF;

    for (i=0; i<256; i++)
    {
        if (i > dim) PORTA = 0x00;
        _delay_us(LED_TIME);
    }
}

int main(void)
{
    DDRA = 0xFF;

    int dim = 0;
    int direction = 1;
    while (1) {
        turn_on_LED_in_PWM_manner(dim);

        dim += direction;

        if (dim == 0) direction = 1;
        if (dim == 255) direction = -1;
    }

    return 0;
}
```